

Zhiheng Zhang

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github.com/zhiheng-zhang-Mera

Melbourne, Australia

Research Profile

Computer Science research engineer focused on reliable agentic software, security boundaries, failure analysis and multi-device systems.

Education

University of Melbourne

Master's degree in Computer Science

Melbourne, Australia

Expected 2026

University of British Columbia

Bachelor of Science in Computer Science

Canada

Completed Summer 2024

Selected Research and Engineering Projects

DS-Hns — *Long-horizon autonomous software-work runtime*

2026–present

github.com/zhiheng-zhang-Mera/DS-Hns

- Building long-horizon repository execution with durable task state, restart/recovery, hardware-aware concurrency and explicit terminal-state evidence.
- Uses failure isolation and recovery-oriented execution to study when a software agent has genuinely completed work rather than merely survived as a process.
- Supports research on coding-agent reliability, autonomous maintenance, AIOps and empirical software-engineering evaluation.

Codex Boss — *Agent orchestration and evidence-based acceptance*

2026–present

github.com/zhiheng-zhang-Mera/Codex-Boss

- Developed a multi-agent workflow platform for planning, execution, critique and adjudication with durable state and evidence-aware acceptance.
- Treats unresolved failure cases and incomplete acceptance conditions as explicit evidence rather than reporting them as solved.
- Supports research on trustworthy agent orchestration, completion claims, auditability and human oversight.

Utopia / Digital City — *Multi-device personal-computing and research platform*

2026–present

github.com/zhiheng-zhang-Mera/Utopia

- Built Android and Web control surfaces around a platform-neutral City protocol with authenticated device/control paths, explicit confirmation and truthful degraded-state handling.
- Current product evidence includes accepted V0.2/V0.3 boundaries and an accepted three-end mesh across two real Windows workers plus an Android control surface with strict target-device routing and cross-host review.
- The research-strengthening programme has accepted experiment registry, trace/provenance, repeated-scenario, fault-injection, and replay/ablation stages; artifact export remained under independent review at the 2026-10-06 snapshot.
- Next research direction: a **Personal Compute Fabric** for heterogeneous-device telemetry, placement/offloading, recovery and explainable cross-device execution; this is a planned extension, not claimed complete.

Research Interests

Software security; program analysis; fuzzing; agent/system vulnerabilities; reliable autonomous software; evidence-driven testing.

Technical and Research Strengths

- End-to-end research engineering: system design, implementation, testing, debugging, controlled experiments and evidence-driven iteration.
- Multi-device and long-running software systems with explicit lifecycle, recovery, routing, observability and acceptance boundaries.
- Git/GitHub-based reproducible development, regression testing, cross-host review and structured preservation of failure evidence.

Current Research Direction

I want to investigate agent and multi-device systems as attack and failure surfaces, using controlled faults and evidence-preserving evaluation rather than only model-level benchmarks.